

DHRITI MITTAL

[LinkedIn](#) • [GitHub](#) • dhritimittal191205@gmail.com • +91 9975717082

Full-Stack & AI Systems Engineer with a proven track record of deploying web applications and specialized deep learning infrastructure. Built a production React/Next.js EdTech platform, a distributed Go-based orchestrator for remote GPU inference, and a LangGraph multi-agent RAG system. Skilled in C++, Python, and TypeScript, with a demonstrated ability to architect both frontend user flows and complex backend machine learning pipelines (Computer Vision, PINNs).

SKILLS

Languages: C++, Python, Go, SQL

Frontend & Product Engineering: React, Next.js, Tailwind CSS, Framer Motion, JavaScript/TypeScript, Component-Driven & Responsive Design, Figma (Wireframing & Prototyping)

AI/ML & RAG Systems: RAG Pipelines, Prompt Engineering, TensorFlow, Deep Learning, Computer Vision, Transfer Learning

Backend & Deployment: Vercel (CI/CD), Git & GitHub, REST API Integration, CMS Data Extraction

EXPERIENCE AND LEADERSHIP

Technical Summer Intern | *LearningShala*

June, 2026 and July, 2026

Working directly with the founder, owning product strategy, UI/UX, frontend architecture, and API integration.

Project – College Discovery & Counselling Platform ([Live Link ↗](#))

- Redesigned core user flows (course discovery, admissions, counselling) for a live university admissions platform in React/Next.js/Tailwind, aiming to reduce drop-off rate through direct weekly iteration with the founder.
- Replaced hardcoded style values with a semantic design-token system under Tailwind v4, minimising new-page styling time and unblocking multi-university theming.
- Migrated 250+ course pages from static data to a live API-driven pipeline, integrating a nested CMS payload structure and resolving multiple Vercel build/config failures to restore reliable CI/CD delivery.

Lead, Operations Department | *VIT Bhopal University — Android Club*

April 2024 – April 2026

- Coordinated 80+ cross-functional team members to run technical events for 1,000+ participants, delivering all events on schedule and within stipulated budgets.
- Owned logistics and stakeholder communication across 25+ events, cutting planning time by 20%.

Co-Lead, Technical Department | *VIT Bhopal University — Matrix Club*

January 2025 – September 2025

- Ran 5 UI/UX and Figma workshops for 25+ members, with 80% reporting increased confidence in design tooling on post-workshop surveys.

PROJECTS

MeshLLM: Distributed Inference Network | *Go, Python, gRPC, Tailscale*

- Built a distributed inference network enabling seamless execution of large language models (Llama 3) across fragmented, remote GPU instances using pipeline parallelism.
- Implemented a Go-based orchestration server handling distributed node heartbeats and API key issuance via gRPC and HTTP over reverse proxies.
- Optimised data-plane networking by establishing a Tailscale mesh, allowing gigabytes of model weights and layer activations to stream directly Peer-to-Peer without bottlenecking the control plane.

RepoSearch AI: Autonomous Codebase Analyst | *LangChain, FastAPI, ChromaDB, Multi-Agent Systems, RAG*

- Engineered a multi-agent AI system using LangChain and LangGraph to autonomously index, analyze, and chat with GitHub repositories, featuring an orchestrator that routes queries to 6 task-specialized agents (e.g., bug hunting, refactoring).
- Architected a hybrid RAG pipeline combining BM25, MMR, and cross-encoder re-ranking with language-aware code chunking, significantly improving context retrieval accuracy across complex, multi-language codebases.
- Built a full-stack application with FastAPI and Next.js, implementing real-time SSE streaming for transparent agent reasoning and supporting privacy-preserving offline analysis via Ollama.

Physics-Informed Projectile Prediction Model | *Ongoing*

- Architecting a hybrid Physics-Informed Neural Network (PINN) and CNN to forecast complex projectile motions, embedding Newtonian kinematic constraints directly into the loss function to ensure physically viable predictions.
- Designing a CNN backbone in PyTorch to extract spatial features from initial flight trajectory data, passing latent representations to a PINN to predict long-horizon trajectories under varying atmospheric drag conditions.
- Optimizing a custom physics-guided loss function to penalize deviations from physical laws, targeting an 90% reduction in long-term prediction drift compared to purely data-driven baseline models.

EDUCATION

B.Tech in Computer Science and Engineering (Core) VIT Bhopal University, Madhya Pradesh	2023 – Present
CGPA: 8.29	
Class 12th (CBSE) Viraj Shri Ram Centennial School, Maharashtra	2023
Percentage: 87.2%	
Class 10th (CBSE) Viraj Shri Ram Centennial School, Maharashtra	2021
Percentage: 99.6%	

CERTIFICATIONS

- **Adobe UI/UX Certification:** User Research, Information Architecture, Wireframing & Prototyping (Figma, Adobe Creative Cloud).
- **Introduction to Machine Learning (NPTEL):** Core ML algorithms, supervised/unsupervised learning, and applications.
- **Marketing Analytics (NPTEL):** Data-driven marketing strategy, customer segmentation, and predictive modelling.
- **The Bits and Bytes of Computer Networking (Google, Coursera):** Network architecture, cloud computing, and the physical layer of networking.
- **ServiceNow Virtual Internship Program (AICTE/SmartBridge):** Administration Fundamentals, Agentic AI, Flows, ATF Essentials.